

# Exact Primal-Dual Certificates for a Two-Bidder, Two-Item Auction

Jiarui Fang

Boston University, Boston, Massachusetts, USA

baymin@bu.edu

ORCID: 0009-0006-9100-0445

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## Abstract

Multidimensional incentives couple allocations across reports and across bidders. For two additive bidders and two items with independent uniform values, we construct a randomized auction satisfying dominant-strategy incentive compatibility (DSIC), ex-post individual rationality (IR) and joint feasibility at every report. Its revenue lies between 0.876514341027 and 0.876514355425. A common capacity-price certificate, a mathematical charge on allocation used to bound every admissible auction's revenue, gives an unrestricted upper bound 0.8823515854888. The gap is below 0.005837245, hence below 0.01, and the auction earns more than 99.3384% of optimal revenue. A global reserve transformation moves both bidders' menus together, transfers ownership on a region of positive measure, and yields an exact sixteen-face revenue formula. Its incentive-preservation argument is independent of the value distribution, and its positive revenue derivative identifies a profitable variation for the specified seed. The upper certificate combines conditional bounds under one capacity price, accounting for the other bidder's virtual value and for overlapping incentive constraints. Continuous computational certificates separate price reductions from integration remainders and state their implementation and arithmetic trust boundaries. The unrestricted optimum and a matching mechanism remain open.

## 1 Introduction

Optimal auction design is completely understood for a single item under the classical regularity conditions of Myerson [6]. With several items, bidder types become multidimensional and dominant-strategy incentive compatibility links allocations across all reports. Even the canonical case of two additive bidders, two items and independent uniform values lacks an exact optimal mechanism [4, 8].

The difficulty is joint: changing one bidder's menu changes the capacity and information rents available to the other. A locally profitable price change can violate supply at a different profile. Conversely, a conditional revenue bound supports the whole auction only if the same capacity price controls both bidders. We construct a feasible mechanism and an unrestricted upper certificate that address these two compatibility requirements directly.

The mechanism applies a common reserve transformation to an explicitly specified system of three conditional menus. It selects both allocations at the same transformed profile, preserving joint supply. Nonnegative allocations and exclusion of zero-valued items at every selected tie preserve DSIC and individual rationality. The transformation transfers ownership on a region of positive measure. Its exact revenue includes all sixteen faces of the transformed distribution, because the reserve creates atoms at zero.

The upper certificate combines a radial revenue identity, a tangent stream field, conditional screening supports and nonnegative translated incentive inequalities. Conditional replacements

explicitly pay the opposing bidder’s virtual-value excess. Corrections on shared endpoint cells are summed before taking the common itemwise maximum. This yields a smaller valid capacity price without imposing a finite-menu restriction on competitors.

The resulting theorem gives an explicit mechanism and certified endpoints for the full continuous randomized DSIC problem. The incentive-preservation argument applies to any Borel seed with bounded payments satisfying its stated hypotheses on this report domain; uniformity is needed for the displayed face revenue formula. The support identities and shared-cell accounting are analytic tools, while the chosen menus, fields and quantitative savings are specific to this instance. We do not claim general convergence or identify an optimal auction.

## 1.1 Relation to prior work

Computational affine-maximizer searches provide transparent deterministic benchmarks [8]. GemNet constructs fully strategyproof menu mechanisms through a compatibility transformation [9]. Jiang, Parkes and Wang report approximately 0.876 revenue for that benchmark and a strict  $50 \times 50$  continuous uniform upper certificate of 0.8919 [4]. We improve the printed upper certificate and give an independently certified revenue enclosure for our explicitly specified auction. The rounded GemNet report does not establish a ranking against its unrounded revenue; its mechanism and the external dual array are not replayed here.

Rochet’s cyclic-monotonicity characterization turns DSIC into convexity of truthful utility and identifies allocation vectors with subgradients [7]. Continuous duality for the one-bidder multi-item problem has been developed through transport and partial-derivative formulations [2, 3]. These results motivate integration-by-parts certificates but do not directly enforce item supply across several bidders.

Multi-bidder dual formulations also predate the present work. Finite-type flow-induced virtual values arise in unified Bayesian mechanism-design duality [1]; Zuo considers both Bayesian and dominant- strategy formulations [10]; and Beckmann-style transport gives a continuous strong-duality formulation for multi-item, multi-bidder BIC auctions in interim space [5]. The latter problem uses weaker incentive and individual-rationality requirements than the pointwise DSIC and ex-post IR requirements studied here.

Most directly, Jiang, Parkes, and Wang construct nonnegative, flow-conserving DSIC deviation kernels, derive continuous weak duality, and lift coarse uniform-grid certificates to continuous type spaces [4]. Our proof shares the same high-level weak-duality architecture: conservation eliminates payments, the dual object induces virtual coefficients, and ex-post feasibility bounds virtual surplus itemwise. The representation and certification are different. We use a signed vector field with a distributional radial source plus a divergence-free tangent stream correction, and prove its validity directly against Lipschitz convex utilities. We do not assert that every such stream field is generated by a nonnegative deviation kernel, or conversely. The contribution is an explicit rational witness for this canonical DSIC instance, an exact Bernstein certificate for its continuous objective, and explicit conditional capacity supports that guide feasible two-bidder reallocations. The convex-mixture concavity of residual value is standard; the new theorem content is the displayed residual-specific measures, their full-class slack identities, and the resulting complete primal variations. We make no first-in-literature priority claim for the general use of convex potentials, transport, or complementary slackness.

## 2 Model and certified bounds

Write a report profile as

$$(x, y, z, w) = (v_{11}, v_{12}, v_{21}, v_{22}) \in [0, 1]^4.$$

The two bidders are risk-neutral and additive; the two items are indivisible. The four coordinates are independent and uniform. A direct mechanism specifies jointly measurable expected

allocation probabilities  $x_{ij} : [0, 1]^4 \rightarrow [0, 1]$  and expected payments  $p_i : [0, 1]^4 \rightarrow \mathbb{R}$ , with  $p_i \in L^1([0, 1]^4)$ . At every report profile, itemwise feasibility requires  $x_{1j} + x_{2j} \leq 1$ . These marginals can be realized by assigning each item to bidder 1, bidder 2, or the seller with the corresponding probabilities. DSIC and ex-post IR are required at every type and opponent report, with utilities averaged over the mechanism's internal randomization. Let OPT be the supremum of expected revenue over this class. Our constructed mechanisms are Borel. The upper argument also covers joint Lebesgue measurability with integrable payments: on every opponent fiber, DSIC supplies a convex Lipschitz utility and the selected allocation equals its gradient off an own-type null set; Fubini supplies the integrated inequality. No pointwise Borel-representative repair is asserted. The measurable origin trace is justified in Appendix B. Residual functions are kept pointwise, not identified modulo area-null sets.

**Theorem 1** (Certified revenue bracket). *The complete mechanism in Proposition 5 has exact revenue  $R_\tau$  given by (4), with*

$$0.876514341027 < R_\tau < 0.876514355425.$$

*For the unrestricted randomized class,*

$$R_\tau \leq \text{OPT} \leq U < 0.882351585489, \quad U - R_\tau < 0.005837245 < \frac{1}{100},$$

*where the rational  $U$  is specified by (10). Moreover,*

$$R_\tau / \text{OPT} \geq R_\tau / U > 0.993384.$$

*The mechanism therefore earns more than 99.3384% of unrestricted optimal revenue. Its revenue enclosure has width less than  $1.44 \times 10^{-8}$ . Full outward-rounded endpoints appear in Appendix A.6; the exact revenue is defined by the face integrals, not a decimal endpoint. This paper neither identifies the optimal revenue nor specifies an optimal mechanism.*

The proof has two independent parts. Section 3 specifies the mechanism, proves its pointwise properties and derives its revenue. Section 4 constructs a common capacity price valid for every admissible competitor. Appendices A–C supply the menu compatibility cases, analytic field proofs and exact integration components. The displayed interval certifies the stated mechanism's revenue; its midpoint is not an exact revenue value.

### 3 A complete mechanism and its exact revenue

We first prove the reserve transformation for a general seed satisfying explicit hypotheses, then give the seed menus and the selected parameter. Together these define the final mechanism at every report.

#### 3.1 A distribution-independent reserve transformation

Let  $M_*$  be any Borel, jointly feasible, pointwise DSIC and ex-post IR mechanism on  $[0, 1]^4$ , with allocations in  $[0, 1]^2$  and bounded expected payments. Assume that its selected allocation excludes zero-valued items,

$$z_{ij} = 0 \implies x_{ij}^*(z) = 0 \quad \text{at every profile, including ties.}$$

Borel measurability and bounded payments ensure that clamping onto boundary faces preserves measurability and integrability. The incentive argument itself uses no distributional assumption

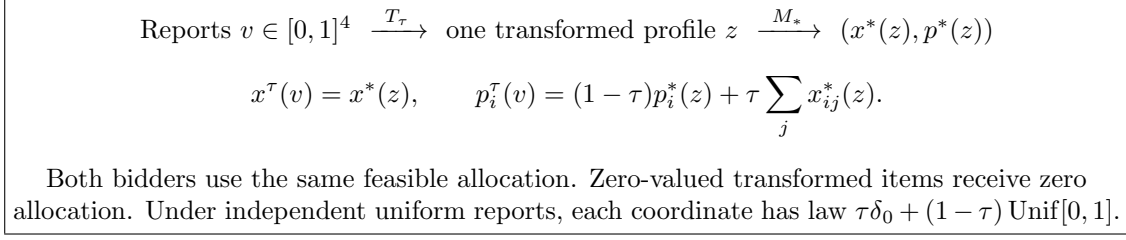


Figure 1: One common report transformation preserves supply and creates the boundary atoms used in the exact revenue calculation. The seed is fully specified by the three menus in Section 3.2.

and no finite allocation range. For  $0 \leq \tau < 1$  put  $s_\tau = 1 - \tau$  and  $T_\tau(t) = \max\{(t - \tau)/s_\tau, 0\}$ , coordinatewise on the full profile. Define

$$x_{ij}^\tau(v) = x_{ij}^*(T_\tau v), \quad p_i^\tau(v) = s_\tau p_i^*(T_\tau v) + \tau \sum_j x_{ij}^*(T_\tau v). \quad (1)$$

Use the seed's given allocation and payment selection at the transformed profile, including its tie rule. There is no additional allocation gate.

**Proposition 2** (Reserve preservation). *For every  $\tau \in [0, 1)$ , (1) defines a Borel, pointwise DSIC, ex-post IR and jointly feasible mechanism on  $[0, 1]^4$ .*

*Proof.* Fix an arbitrary opponent report, a true type  $v_i$ , and a report  $r_i$ . Its deviation utility is bounded by

$$\begin{aligned} & \sum_j (v_{ij} - \tau) x_{ij}^*(T_\tau r_i, T_\tau v_{-i}) - s_\tau p_i^*(T_\tau r_i, T_\tau v_{-i}) \\ & \leq s_\tau [T_\tau v_i \cdot x_i^*(T_\tau r_i, T_\tau v_{-i}) - p_i^*(T_\tau r_i, T_\tau v_{-i})] \\ & \leq s_\tau u_i^*(T_\tau v). \end{aligned}$$

The first inequality uses nonnegative allocations; the second is DSIC of  $M_*$ . At truth the first inequality is equality by the zero-value-exclusion hypothesis, including  $v_{ij} = \tau$  and  $v_{ij} < \tau$ . Hence truthful utility is  $s_\tau u_i^*(T_\tau v) \geq 0$ . Both allocations come from the same feasible transformed profile, so capacity holds pointwise. The map is continuous and the selected base rule is Borel. Bounded seed payments keep transformed payments bounded, so integrability also holds. All assertions include exceptional reports and ties and use the expected-utility convention of Section 2.  $\square$

### 3.2 The seed menu system

Put

$$\begin{aligned} A &= \frac{2}{3}, \quad d = \frac{1}{2}, \quad c = \frac{157}{500}, \quad q = \frac{93}{500}, \quad \nu = \frac{q^2}{2}, \quad a = A + \nu, \\ b &= a + c = \frac{1496947}{1500000}, \quad s = a + d = \frac{1775947}{1500000}, \quad \lambda = \frac{4}{5}, \quad \bar{u} = \frac{3421}{10000}. \end{aligned}$$

The symbol  $\bar{u}$  denotes a threshold parameter, not utility. Define

$$g(r) = \lambda(\bar{u} - r) \mathbf{1}_{[c, \bar{u}]}(r), \quad h(r) = r + q + g(r), \quad J = g(c) = \frac{281}{12500}. \quad (2)$$



### 3.3 The specified transformed mechanism

We apply Proposition 2 to the complete seed of Section 3.2.

**Lemma 4** (Zero-value exclusion). *At every profile  $z$ , the selected allocation of  $M_*$  satisfies  $z_{ij} = 0 \Rightarrow x_{ij}^*(z) = 0$ , including all ties.*

*Proof.* In a base menu each singleton costs at least  $1/2$ , and each bundle upgrade costs at least  $c = 157/500$ . In a free  $Q$  menu both upgrades cost at least  $b_0 - A > 0$ . In a constrained  $Q$  menu,  $k \geq c$  and  $C \geq 5/6 + 3k^2/4$ , with safe price  $B = \min(A, C - k)$ . These imply positive singleton prices and positive bundle upgrades on the complete certified region. The uniform safe-price and safe-upgrade lower bounds can be taken as  $525947/10^6$  and  $721841/3000000$ . In an  $E$  lottery, positive safe allocation requires safe value at least  $c$ , and positive scarce allocation requires scarce value at least  $A - q - \delta(A) \geq 1364159/3000000 > 0$ . Thus any row allocating a zero-valued item is strictly dominated by an available row deleting that item. The empty row is selected whenever maximum utility is zero; positive ties remain between maximizing rows that have the same exclusion property. The base,  $Q$  and  $E$  cases cover the complete mechanism.  $\square$

**Proposition 5** (Complete specified mechanism). *Use  $\tau = 83/10000$  in (1), with the seed menus and joint tie rule of Section 3.2. This defines a Borel, pointwise DSIC, ex-post IR and jointly feasible mechanism.*

*Proof.* The seed satisfies Proposition 3 and Lemma 4. Its nonnegative payments are at most two by IR. Proposition 2 therefore applies at every report.  $\square$

This change need not retain a subset of the original allocation at the same report. Throughout the closed box of halfwidth  $10^{-4}$  centered at

$$(v_{11}, v_{12}, v_{21}, v_{22}) = (1577/2000, 123/200, 3/5, 4/5),$$

$M_*$  gives both items to bidder 1, whereas  $M_\tau$  gives item 1 to bidder 1 and item 2 to bidder 2. Exact branch inequalities first place the entire box in the stated base-menu regime. All remaining utility differences are affine there; their vertex bounds are at least  $1/1250$  before the change and  $1719/5000000$  after it. The box has volume  $1/625000000000000$ . This is a jointly feasible ownership transfer, not a sampled allocation comparison.

### 3.4 Exact revenue through face moments

For  $S \subseteq \{1, 2, 3, 4\}$  let  $R_S$  and  $N_S$  be the integrals of the total payment and total expected number of allocated items of  $M_*$  on the face where coordinates in  $S$  are zero, with all remaining coordinates independent uniform. Each transformed coordinate has law  $\tau\delta_0 + s_\tau \text{Unif}[0, 1]$ . Independence gives the exact degree-five polynomial

$$R_\tau = R(M_\tau) = \sum_{S \subseteq \{1, 2, 3, 4\}} \tau^{|S|} s_\tau^{4-|S|} (s_\tau R_S + \tau N_S), \quad \tau = 83/10000. \quad (4)$$

This formula, together with the complete menus, is an exact revenue characterization. The numerical value is certified by an interval; neither its midpoint nor an endpoint is asserted to be the exact revenue. Boundary faces cannot be dropped: they have positive probability after transformation, even though they had zero four-dimensional volume before it.

For clarity, the sums over faces of each codimension are

$$\begin{aligned} (\mathcal{R}_0, \dots, \mathcal{R}_4) &= (R_*, 4R_3, 2(R_{\text{SB}} + R_{\text{same}} + R_{\text{cross}}), 4R_1, 0), \\ (\mathcal{N}_0, \dots, \mathcal{N}_4) &= (N_4, 4N_3, 2(N_{\text{SB}} + N_{\text{same}} + N_{\text{cross}}), 4N_1, 0). \end{aligned}$$

Here  $R_3, N_3$  refer to one fixed zero coordinate;  $R_1, N_1$  to one remaining nonzero coordinate. Of the two-dimensional faces, one bidder alone yields  $R_{\text{SB}} = 4/9 + 2\sqrt{2}/27$  and  $N_{\text{SB}} = (8 + 2\sqrt{2})/9$ ; the same-item face yields  $R_{\text{same}} = 31/81$ ,  $N_{\text{same}} = 5/9$ ; and the cross-item face retains the two opposing singleton menus. Also  $R_1 = 2/9$ ,  $N_1 = 1/3$ . The remaining coefficients reduce to conditional-menu integrals as follows. Let  $\mathcal{T} = \{(t, \rho) : 0 \leq \rho \leq t \leq 1\}$ . Write  $r_c(t, \rho)$  and  $n_c(t, \rho)$  for the expected payment and expected item count of the single bidder facing  $(t, \rho)$  under  $M_*$ , including its lottery options, with its own two coordinates uniform. Let  $P(t, \rho)$  and  $B(t, \rho)$  be the scarce and safe singleton prices clipped at one. With  $g_R = P(1 - P) + B(1 - B)$  and  $g_N = 2 - P - B$ , the exact reductions are

$$\begin{aligned} N_4 &= 4 \int_{\mathcal{T}} n_c(t, \rho) dt d\rho, \\ R_3 &= \int_0^1 r_c(t, 0) dt + \int_{\mathcal{T}} g_R(t, \rho) dt d\rho, \\ N_3 &= \int_0^1 n_c(t, 0) dt + \int_{\mathcal{T}} g_N(t, \rho) dt d\rho, \\ R_{\text{cross}} &= 2 \int_0^1 B(t, 0)(1 - B(t, 0)) dt, \\ N_{\text{cross}} &= 2 \int_0^1 (1 - B(t, 0)) dt. \end{aligned} \tag{5}$$

On a one-zero face, the full two-coordinate bidder gives the line integral, and the other bidder's singleton sales under the two opponent orientations give  $g_R$  and  $g_N$ . On a cross face only the safe singleton threshold remains. Zero-coordinate exclusion removes any competing bundle or lottery allocating the zero-valued item. The formulas and polynomial branch integrands are also specified in `verifier/primal_face_integrals.py` in the certificate package.

The pointwise tie rule need not be symmetric. The face integrals nevertheless have the required symmetries: zero-coordinate exclusion handles fixed zero coordinates, ties involving positive free coordinates lie on null own-type menu interfaces, and a bidder whose two coordinates are fixed at zero selects empty. These facts justify the face symmetries without changing the selected pointwise rule.

The verification partitions the opponent triangle into 296 fully resolved strips and 15 wholly enclosed unresolved strips. Polynomial branch conditions are certified over entire cells before exact integration; unresolved strips receive valid global payment and allocation bounds. The single-buyer radical is bracketed rationally, with a conservative  $10^{-12}$  face error allowance justified by the utility revenue identity. An independent standard-library implementation reconstructs 1,372 polynomial cells and checks 63,184 exact Bernstein predicates before integrating them. It agrees on every face enclosure and on the full 16-face expansion. It shares the analytic menu formulas and enclosure principle, but imports no source calculator.

The same exact identity gives a global admissible first variation,

$$\left. \frac{d}{d\tau} R(M_\tau) \right|_{0+} = N_4 + 4R_3 - 5R_* \in [0.0120503811119742, 0.0120521625636254].$$

Thus this necessary optimality condition fails at  $M_*$ . The selected finite change at  $\tau = 83/10000$  gains more than  $3/62500$  over the seed, as follows from the certified revenue and seed integral. Appendix A.6 certifies the unique maximum within this fixed-seed family and explains the selected rational parameter. It does not identify an unrestricted optimal auction.

## 4 A common capacity certificate for all randomized mechanisms

### 4.1 Residual capacity and a common upper certificate

Fixing one bidder's complete mechanism leaves a constrained screening problem for the other. This gives a useful interpretation of a capacity price. For a jointly measurable capacity function  $r : [0, 1]^4 \rightarrow [0, 1]^2$ , let

$$\mathcal{V}(r) = \sup\{\text{Rev}_2(M_2) : M_2 \text{ is pointwise DSIC/IR and } x_2 \leq r\}.$$

The mechanisms in this definition remain jointly measurable, with integrable payments. Grouping feasible pairs by their first component gives exactly

$$\text{OPT} = \sup_{M_1}\{\text{Rev}_1(M_1) + \mathcal{V}(1 - x_1)\}.$$

This identity does not assert existence of a best response or invoke a measurable selection of pointwise optimizers.

Write  $a_i(t; w) = x_i(t, w)$  for bidder  $i$ 's allocation vector, with own report  $t$  and opponent report  $w$ . For a measurable, integrable payment slice, put  $R_i(w) = \int_{[0,1]^2} p_i(t, w) dt$ , so that  $\text{Rev}_i(M_i) = \int R_i(w) dw$ . Such slices exist for almost every  $w$  in the stated jointly measurable class. On a fixed fiber define

$$R(u, a) = \int_{[0,1]^2} [t \cdot a(t) - u(t)] dt,$$

$$V_w(s) = \sup\{R(u, a) : (u, a) \text{ admissible on the fiber, } a \leq s\}.$$

Here admissible means  $u \geq 0$  is convex,  $a(t) \in \partial u(t)$  and  $0 \leq a \leq 1$  at every report, with measurable allocation and integrable payment  $t \cdot a(t) - u(t)$ . Thus  $V_w$  is a conditional screening value;  $\mathcal{V}$  remains the value over complete jointly measurable mechanisms. No equality between  $\mathcal{V}$  and an integral of fiberwise suprema is assumed.

**Proposition 6** (Residual value and supporting measures). *The functional  $\mathcal{V}$  is nondecreasing and concave. On a fixed opponent fiber  $w$ , suppose a finite nonnegative capacity-price measure  $\pi$  satisfies  $R(u, a) \leq \langle \pi, a \rangle$  for every admissible conditional mechanism, and a feasible candidate at  $r^*$  attains  $R^* = \langle \pi, r^* \rangle$ . Then that candidate solves the full randomized conditional problem and, for every residual for which the pairing is defined,*

$$V_w(r) \leq V_w(r^*) + \langle \pi, r - r^* \rangle.$$

*Proof.* Monotonicity is immediate. Mix any two admissible mechanisms using one exogenous coin with constant probability  $\theta$ . All pointwise incentive, IR and capacity inequalities are preserved, as are measurability and integrability. Apply this to arbitrarily near-optimal competitors and pass to the supremum. For the second assertion,  $R \leq \langle \pi, a \rangle \leq \langle \pi, r \rangle$ , and equality at  $r^*$  gives the claim. No own-report-dependent mixture is used.  $\square$

The common-certificate requirement is stronger than conditional optimality at a single residual. On  $\Omega = [0, 1]^4$ , suppose nonnegative finite capacity measures  $\Pi_j$  have well-defined pairings and satisfy

$$\text{Rev}_i(M_i) \leq \sum_j \int_{\Omega} x_{ij} d\Pi_j$$

for every complete DSIC/IR mechanism of either bidder. Joint feasibility then gives  $\text{Rev}_1(M_1) + \text{Rev}_2(M_2) \leq \sum_j \Pi_j(\Omega)$ . Equality requires both revenue inequalities and all priced capacity constraints to bind at the same joint allocation. The construction below uses integrable densities, so these pairings cover the measurable class of Section 2.



## 4.2 Revenue fields and the certificate components

For each bidder, the integrable radial field

$$F^0(x, y) = \frac{3r^2 - 1}{2r^2}(x, y), \quad r = \max(x, y), \quad F^0(0, 0) = 0,$$

and the divergence-free tangent correction  $G_i$  define  $\phi_i(v) = F^0(v_i) + G_i(v)$ . The rational witness  $G_i$  is specified in Appendix B. The resulting conditional identity is

$$R_i(v_{-i}) = \int_{[0,1]^2} \phi_i(v_i; v_{-i}) \cdot a_i(v_i; v_{-i}) dv_i - u_i(0; v_{-i}).$$

Appendix B proves the identity for Lipschitz convex utilities and specifies every rational stream coefficient. A second nonnegative field  $\Psi$  satisfies  $\langle \Psi, a \rangle - R = 3 \int_{D_0} u \geq 0$  for every conditional DSIC/IR mechanism. Its formula, no-sale region  $D_0$  and the square  $W_0 = [0, 43/100]^2$  are given in Appendix C.

Write  $B$  for the integral of the stream maximum,  $B_{20}$  for its certified Bernstein majorant and  $D$  for the reduction obtained by switching to  $\Psi$  on opposing reports in  $W_0$ . The exact lower bound  $\Delta_{1024} \leq D$  and four compatible incentive cycles define the reference accounting constant

$$U_{\text{ref}} = B_{20} - \Delta_{1024} - 81/10^{10}.$$

This constant is used only to assemble the final upper  $U$ . The certificate below includes all support replacements and all incentive corrections at once.

## 4.3 Enlarged conditional support and coupled incentive flows

We specify the common measure and certify its total mass. The essential change is to account for the opposing stream field when enlarging the replacement region; its sign need not remain negative.

Put  $J = (43/100, 1/2] \times [0, 7/20]$  and  $D^\sharp = W_0 \cup J \cup J^{\text{swap}}$ . The reusable support (12) and its item swap both vanish on  $D^\sharp \subset D_0$ . Let  $\sigma(t_1, t_2) = (t_2, t_1)$  and  $J^{\text{swap}} = \sigma J$ . Set  $o_i(w) = \sigma$  on  $J^{\text{swap}}$  and  $o_i(w) = \text{id}$  elsewhere, with  $\Psi^{\text{id}} = \Psi$  and  $\Psi^\sigma(t) = \sigma \Psi(\sigma t)$ . Thus  $o_i$  depends only on the opponent report and the density is expressed in physical item coordinates. The no-sale set  $D_0$  is invariant under  $\sigma$ . For  $\theta_i = \mathbf{1}_{\{v_{-i} \in D^\sharp\}}$ , the conditional identity, valid for almost every opponent report in the stated jointly measurable class, is

$$R_i(v_{-i}) = \int [(1 - \theta_i)\phi_i + \theta_i\Psi^{o_i}] \cdot a_i - (1 - \theta_i)u_i(0; v_{-i}) - 3\theta_i \int_{D_0} u_i. \quad (6)$$

The mixing coefficient depends only on the opponent; consequently the identity introduces no own-type interface derivative. All source terms are nonnegative under IR, including for unnormalized utilities.

**Actual IC inequalities on shared endpoint cells.** Combine the four sparse cycles with a coupled system of 129 nonnegative two-way translated IC columns. Column  $\ell$  fixes bidder  $i$ , an opponent rectangle  $J_\ell$ , a common own-offset rectangle  $K_\ell$ , two own centers  $A_\ell, B_\ell$  and a density  $\lambda_\ell \geq 0$ . With  $d_\ell = A_\ell - B_\ell$ , the two truthful inequalities give

$$K_\ell(a) = \lambda_\ell \int_{K_\ell \times J_\ell} d_\ell \cdot [a_i(A_\ell + s, w) - a_i(B_\ell + s, w)] ds dw \geq 0.$$

The induced allocation coefficients are  $+\lambda_\ell d_\ell$  on the first box and  $-\lambda_\ell d_\ell$  on the second. Densities refer to Lebesgue measure, not normalized probabilities. Every column admits this direct IC proof; no Hessian-density product or singular positive-semidefinite representation is needed here.

In the base chart the system has 159 distinct endpoint cells, of which 54 are shared. On a cell  $Q_c$ , sum every column correction before taking the itemwise maximum. If  $L_{cjk} \leq \max_l \phi_{lj} - \phi_{kj}$  for the old alternatives  $k = 0, 1, 2$  (seller included), then

$$\max_k (\phi_{kj} + \delta_{cjk}) - \max_k \phi_{kj} \leq \max_k (\delta_{cjk} - L_{cjk}).$$

Exact whole-cell bounds therefore certify

$$G_{\text{master}} = -2 \sum_c |Q_c| \sum_j \max_k (\delta_{cjk} - L_{cjk}) > 0.$$

The factor two is simultaneous item exchange only. The original and independent master verifiers reconstruct the cells, shared incidences and aggregate epigraph. The rational centers, widths, columns and amplitudes are given in `master_certificate.json` within the portable source closure.

A further translated IC construction uses 64 selected box pairs and four symmetry copies, hence 512 endpoint boxes. Their entire old and new virtual winner regions are verified before integration, giving

$$G_{\text{BB}} = \frac{4786305147}{17179869184000000}. \quad (7)$$

All added supports avoid profiles with either bidder in  $D^\sharp$ ; the sparse, master and BB supports are mutually disjoint except for the explicitly aggregated master endpoints. The independent audit checks all 130,816 pairwise separations among the BB endpoint boxes. The correction is a nonnegative sum of actual IC inequalities regardless of whether those inequalities bind at the proposed mechanism.

Let  $\bar{\phi}$  be the full radial-plus-stream field with these sparse, master and BB corrections. Define, in physical item coordinates,

$$\alpha_i = (1 - \theta_i) \bar{\phi}_i + \theta_i \Psi^{o_i}, \quad \Pi_j = \max(0, \alpha_{1j}, \alpha_{2j}). \quad (8)$$

Disjointness ensures that the opponent switch does not cut any IC flow. Integrating (6) and adding the nonnegative IC terms proves  $\text{Rev}_i(M_i) \leq \langle \Pi, a_i \rangle$  for each complete DSIC/IR mechanism. Thus both unrestricted charged values are zero (empty attains zero), and joint capacity gives  $\text{OPT} \leq \int \sum_j \Pi_j$ . This proof applies to arbitrary randomized allocations in the stated measurable class; it places no finite-menu restriction on competitors.

**Paying the opposing field on the enlarged region.** For  $w \in J$ , write  $a = \phi_j(v; w)$ ,  $b = \phi_j(w; v)$  and  $p = \Psi_j(v) \geq 0$ . On a single-replacement profile,

$$\max(0, a, b) - \max(p, b) \geq a_+ - p - (b - p)_+,$$

because  $\max(p, b) = p + (b - p)_+$  and the old maximum is at least  $a_+$ . On double-replacement profiles the two directed bounds cancel, and the old nonnegative price covers them. On  $W_0$  the extra directed contribution is nonpositive and remains a valid lower bound. Consequently the complete splice improvement satisfies

$$G_{\text{splice}} \geq 4 \left[ L_- - \frac{49}{2000} R_{\text{SB},+} - E_+ \right], \quad (9)$$

where  $L = \int_J \int \sum_j (\phi_j(v; w))_+$  and  $E = \int_J \int \sum_j (\phi_j(w; v) - \Psi_j(v))_+$ . The subscripts denote directed bounds, so the right side is rational. The factor four counts the two bidder directions and the two oriented added rectangles  $J, J^{\text{swap}}$ . The item sum is already inside  $L$  and  $E$ , so it

contributes no further multiplicity. The low-square and double-replacement cases are expanded in Lemma 10 of Appendix C.1. The certificate obtains

$$L_- = \frac{119448693432877}{8796093022208000}, \quad E_+ < 0.000011750019244,$$

yielding a guaranteed reduction exceeding 0.00045027837569. Opposing excess and overlap are both included.

The  $L$  calculation uses continuous Jensen bounds on 49 opponent cells and  $128^2$  own-chart cells. The  $E$  certificate covers 294 roots with 12,000 splits and 12,294 leaves. Its interval arithmetic starts with exact rational input brackets and expands every binary64 operation outward using `nextafter`; final dyadic bounds are accumulated with integers. IEEE binary64 round-to-nearest semantics and NumPy’s `nextafter` are therefore additional trusted assumptions. The full tree was freshly replayed. The separate splice audit rebuilds the identity and selected exact checks but reuses the primary tree implementation; it is not a second independently implemented full tree.

**The exact endpoint and its remaining slack.** The reference enclosure remainder is  $E_* = (B_{20} - B) + (D - \Delta_{1024}) \geq 0$ . A fresh fixed-point traversal at depths (20, 20, 26) recovers a rational lower bound  $E_- \leq E_*$ , approximately 0.00012090110466538207. This contracts numerical enclosure; it is not another decrease in the actual measure. With the rational lower bound in (9) denoted by  $G_{\text{splice},-}$ , set

$$U = U_{\text{ref}} - E_- - G_{\text{master}} - G_{\text{splice},-} - G_{\text{BB}}. \quad (10)$$

Every rational is exported in the final manifest and checked by separate exact endpoint assembly. Equation (10) contains every term used in the final upper, with no additional unlisted gain.

| Certified component   | Approximate size           | Mathematical role                                |
|-----------------------|----------------------------|--------------------------------------------------|
| $E_-$                 | $1.2090110 \times 10^{-4}$ | Recovered integration remainder                  |
| $G_{\text{master}}$   | $9.6899332 \times 10^{-9}$ | Common-price decrease from aggregated IC columns |
| $G_{\text{splice},-}$ | $4.5027838 \times 10^{-4}$ | Common-price decrease paying opposing excess     |
| $G_{\text{BB}}$       | $2.7859963 \times 10^{-7}$ | Common-price decrease from translated IC pairs   |
| $U_{\text{ref}} - U$  | $5.7146777 \times 10^{-4}$ | Exact sum of the four subtractions               |

Table 1: Components assembling the final upper from the reference majorant  $U_{\text{ref}}$ . The table uses descriptive rounded decimals; the exact manifest rationals satisfy (10). The first row contracts numerical enclosure. The other rows certify decreases in the actual common measure, with the enlarged splice giving the largest change.

For any candidate,  $\int \Pi - R$  separates into the source terms in (6), nonnegative IC residuals, unused priced capacity, and virtual-allocation mismatch. The reported  $U$  also includes nonnegative integration remainders. At  $M_\tau$  the source terms vanish:  $T_\tau \leq \text{id}$  keeps the relevant no-sale types at zero utility and the origin utility is zero. This equality is not conditional optimality of its residual menus. The other terms have not been made zero or fully localized for  $M_\tau$ . Conditional screening at the seed residuals does not establish equality at the transformed mechanism. A matching certificate remains open.

## 5 Reproducibility and trusted computations

The certificate package specifies the full continuous mechanism, all rational coefficients and the exact component assembly. Discovery optimizations are not proof inputs. Polynomial cells are

accepted only after inequalities are certified over their whole domains; finite report checks serve implementation validation and do not establish continuous incentive compatibility.

The mechanism’s DSIC, IR and feasibility follow from the all-real arguments in Section 3 and Appendix A. Its seed revenue is independently reconstructed by polynomial/simplex integration, polygon boundary integrals and vertical slicing. A separate implementation, importing no source calculator, reconstructs every face enclosure used by the reserve formula. These calculations share the analytic menu specification.

The upper verification reconstructs the stream polynomials and complete Bernstein coverage. The  $B_{20}$  majorant has two complete implementations. The  $\Delta_{1024}$  subtraction has one full accumulation, with a separate reconstruction of averaged coefficients, classifications through size 64 and selected cell integrals. The master system has a separate complete replay of its shared cells and aggregate bounds. The opposing-excess tree has a full directed-arithmetic replay and bounded independent cross-checks, rather than a second independently implemented full tree.

Rational polynomial arithmetic, integer accumulation and directed error bounds support the numerical enclosures. The opposing-excess traversal also assumes IEEE binary64 round-to-nearest arithmetic and outward expansion by NumPy’s `nextafter`. The trusted components are the written analytic arguments, the stated arithmetic semantics and the verification programs. This is a computer-assisted proof package, not a proof-assistant formalization.

The package README maps the manuscript’s mathematical components to their machine-readable files and gives a single reproduction command. It checks the certificate dependencies, exact endpoint assembly, manuscript consistency, clean compilation and complete SHA-256 coverage. Existing source directory names are implementation identifiers; they do not specify an additional auction or change the final theorem. Hashes establish file identity, separately from analytic validity and computational replay.

The final parameter and endpoint record is `certificate/reserve_parameter/manifest.json`, identified by SHA-256

```
bf6da386fa6309eab3a4c8379b5f91f8
4d087322a69fb7a9fbcc7221bbb7bdb1
```

This content identifier fixes the selected reserve, face-input bindings and upper assembly independently of the repository’s mutable default branch. The accompanying `SHA256SUMS` covers the complete release files.

## 6 Discussion and open problem

An explicit jointly feasible mechanism earns more than 99.3384% of optimal revenue, and its certified gap to an unrestricted upper is below 0.005837245. The construction prices both physical scarcity and the incentive restrictions that accompany a change in capacity. The remaining gap is mathematical; it is not a sampling error or an optimization tolerance.

The reserve transformation provides a globally admissible revenue variation. Its validity includes exceptional reports and ties, and its revenue includes the positive-probability faces created by clamping. The positive derivative at the seed and the separately certified finite gain show what this variation accomplishes. Neither result proves optimality of the chosen reserve or of the transformed mechanism.

On the upper side, opposing-field charges and shared-cell aggregation make conditional replacements and incentive flows compatible in one common price. Zero charged screening values alone do not supply an equality auction: the same allocation must also use priced capacity and satisfy the virtual and incentive equalities. Numerical enclosure remainders are kept separate from these mechanism slacks.

The reusable content consists of the incentive-preserving transformation, exact face accounting and common-price compatibility arguments under their stated hypotheses. The quantitative

mechanism and certificate are specific to the uniform two-bidder, two-item instance. Convex utility, transport duality and complementary slackness are established principles; no first-in-literature or cross-instance performance claim is made. The rounded external GemNet benchmark does not resolve the comparison with its unrounded revenue.

The optimal mechanism and a matching global upper bound remain open. Stationarity in a restricted family and failure to find a certificate in one representation do not establish unrestricted optimality.

## Data, code, and materials availability

The project repository is publicly accessible at

<https://github.com/Baymax-ray/two-bidder-two-item-dsic-certificates>.

The accompanying release archive contains all exact certificate data, deterministic verification code, transcripts, environment information, and SHA-256 manifests needed to reproduce the stated bounds. Manuscript, certificate data, and other textual materials are released under CC BY 4.0. Verification software is separately released under the MIT License. The separate research workspace retains discovery-only experiments; the publication entrypoints use the bundled proof inputs.

## AI-assisted tools declaration

This manuscript was developed with the assistance of OpenAI GPT-5.6 Sol and OpenAI Codex. The tool assisted with mathematical exploration, numerical candidate generation, implementation and debugging of certificate and verification code, and drafting and revision of the manuscript. Separately scoped AI-assisted audits checked the new proofs, source implementations and exact arithmetic; these are internal checks, not external peer review. AI-generated suggestions and numerical search outputs were not treated as proofs; the stated results are supported by the analytic arguments and exact certificates described here. The author retains responsibility for the content, correctness, and final presentation of this work.

## A Pointwise menu compatibility and seed revenue

### A.1 Threshold bounds

We use exactly the parameters and menus of Section 3.2. Put  $K = A - q = 721/1500$ ,  $\rho_* = b - A = 165649/500000$  and  $B_0(k) = C_0(k) - k$ . On constrained  $Q$ ,  $c \leq k \leq K < d$  and

$$d < B_0(K) \leq B_0(k) \leq B_0(c) < A, \quad B_0(K) - h(\rho_*) = \frac{37}{5000000} > 0.$$

The function  $B_0$  decreases on this interval. Outside the fee support,  $t - k \geq q$  gives  $z \geq k + h(\rho)$ . On its support,  $Q$  implies  $k \leq \min(K, b - \rho - q)$ . For  $\rho \leq \rho_*$ , the displayed slack bounds  $B_0(k) - h(\rho)$  below. For  $\rho \geq \rho_*$ , its sufficient lower bound  $B_0(b - \rho - q) - h(\rho)$  has derivative at least  $\lambda - 3K/2 = 79/1000 > 0$ , so the same bound applies. Consequently

$$C = \max(C_0(k), z)$$

on every constrained  $Q$  report. If  $B$  is capped at  $A$ , then  $C = z > b_0$ . This argument includes the fee face and does not assume  $C_0(k) \leq 1$ .

A second bound needed below is

$$C_0(k) - k - d \geq g(k).$$

Off the support it follows from  $B_0(k) > d$ . On  $[c, \bar{u}]$ , the difference has positive derivative  $3k/2 - 1 + \lambda$  and minimum  $212401/3000000$ . Since  $t \leq h(k)$ , this also proves  $C_0(k) \geq t + c$ . The inverse implications  $x < k(t) \Rightarrow h(x) < t$  and  $x \geq k(t) \Rightarrow h(x) \geq t$  hold through the whole plateau; an identity  $h(k(t)) = t$  is not used.

## A.2 Base/base compatibility

Before the fee, the base menus are the pivot-normalized menus of a common nine-outcome affine maximizer. Its costs are zero for empty,  $a$  for either singleton allocated to one bidder,  $b$  for either bidder's bundle, and  $s$  for either split allocation. Selecting a maximizer of total reported value minus cost supplies a jointly feasible pair of menu maximizers. Deleting any zero-utility allocation preserves such a pair with the stated empty rule.

The prices satisfy the strict bundle-discount inequality

$$P_{\text{scarce}} + P_{\text{safe}} - P_B \geq q > 0.$$

If both opponent coordinates are at most  $d$ , the left side is at least  $a - c > q$ . If exactly one exceeds  $d$ , use  $H \geq t - a$  to obtain  $q$ ; if both exceed  $d$ , use  $H \geq z - b$  to obtain  $2q$ .

Adding the same nonnegative fee to safe and bundle preserves a maximizing subset of every uncharged maximizer. Indeed, if safe maximizes, comparison with bundle gives  $x \leq P_B - P_{\text{safe}} < P_{\text{scarce}}$  for the own scarce value  $x$ . The scarce singleton has negative utility, and the bundle-minus-safe comparison is unchanged by the fee, leaving safe or empty as a maximizer. A scarce or empty maximizer remains a maximizer because its price is unchanged. A bundle contains every possible new allocation. Applying this argument to the jointly feasible uncharged pair proves base/base compatibility, including positive ties.

## A.3 Q/base and Q/Q compatibility

Fix a  $Q$  report  $w = (t, \rho)$ , with physical item 1 high, and write the other report as  $v = (x, y)$ . A base singleton costs at least  $d$  and its bundle at least  $b$ . Since  $t + \rho \leq b$ , an own type  $w$  can buy only its high item positively in a base menu; if  $t \leq d$  it must be empty.

For  $y \leq d$ , its item-1 base price is at least  $\max(x, x + y - c)$ . An opposing  $Q$  scarce purchase with positive utility requires  $x > A \geq t$ . A selected opposing bundle has  $x + y \geq C \geq C_0(k) \geq t + c$ . Whenever the  $Q$  bidder consumes item 1, the base high price is therefore at least  $t$ , so the base bidder has no positive utility and selects empty.

For  $y > d$ , the base price is at least  $x + q$ . A positive high purchase by  $w$  would require  $x < t - q \leq K < d < y$ . Thus  $x$  is the low coordinate of the base report and the fee on that physical item gives a price at least  $h(x)$ . The opposing  $Q$  scarce singleton requires  $x \geq A > k$ ; a bundle requires  $x \geq C - B \geq k$  by comparison with safe. In either case  $h(x) \geq t$ , contradicting the positive base purchase. The inequalities include the inverse knots, the fee face and capped bundle rows.

For  $Q/Q$ , each own coordinate is at most  $A$ . Neither a scarce singleton at  $A$  nor a capped safe singleton at  $A$  has positive utility. An uncapped singleton costs more than  $d$ , while a constrained  $Q$  type's low coordinate is below  $d$ . Positive singletons can therefore only be distinct own high items under opposite orientations. Two positive bundles are impossible, because each bundle price is at least the opposing report's total value.

For a possible bundle/safe conflict under opposite orientations, a positive safe buyer with high value  $y$  and opposing low value  $\rho$  requires  $y > B \geq h(\rho)$ . A conflicting bundle in the reverse menu requires  $\rho \geq k(y)$ ; the cap only increases this upgrade threshold. Hence  $h(\rho) \geq y$ , a contradiction. The strictness comes from the safe buyer's positive utility, so a tied bundle/safe comparison on the other side is covered. Same orientations have no positive singleton. Free  $Q$  types cannot buy either singleton positively and cannot buy the opposite constrained safe option positively. Empty at zero completes the tie cases.

#### A.4 Compatibility of lottery menus

For an  $E$  menu put  $L = \delta + \alpha/2$ ,  $Y = c + L$  and  $j = 1 - t/2$ . The factored formula for  $\delta$  gives, throughout  $A < t < T$ ,

$$\beta L = \alpha/2, \quad t + c + \delta - Y = j, \quad c < Y < d < d + \delta < A, \quad d < j < A < t,$$

with  $0 < \beta < 1$  and bundle upgrade threshold  $t - q > c$ .

Consider  $E$  against an uncharged base menu first. An own  $E$  type  $w = (t, \rho)$ ,  $\rho < c$ , cannot buy its low base singleton; its base bundle is dominated by high because  $P_B - P_1 \geq c > \rho$ . Only item 1 can conflict. If the other report  $(x, y)$  has  $y \leq d$ , the base high price is at least  $\max(x, x + y - c)$ . An  $E$  scarce purchase requires  $x \geq t$ , a bundle requires  $x + y \geq t + c + \delta$ , and a lottery requires  $x + \beta(y - c) \geq t$ . For a lottery,  $y \geq c$  implies  $x + y - c \geq t$ , while  $y < c$  implies  $x \geq t$ . Thus every allocation consuming item 1 makes the opposing base bidder empty. If  $y > d$ , the base high price is at least  $x + q$ , the lottery cannot win because  $y > Y$ , and a bundle requires  $x \geq t - q$ . Again the base bidder is empty. The maximizing-subset argument permits the actual base fee without changing the  $E$  allocation.

For  $E/Q$ , an own  $Q$  report has coordinates at most  $A$  and sum at most  $b = a + c$ . Its  $E$  scarce singleton has negative utility. Its lottery utility is at most  $A - t + \beta(a - A)$ : for safe value  $y \geq c$ , write the lottery value as  $(1 - \beta)x + \beta(x + y - c)$ ; for  $y < c$ , scarce dominates lottery. Now

$$(t - A)/\beta = t - A + 2\delta/3$$

is strictly increasing, with derivative at least  $d + c > 0$  and infimum  $q^2/2 = a - A$  at  $A$ . The lottery utility is strictly negative on the open  $E$  interval. Bundle utility is at most  $a - t - \delta < 0$ , because  $t - A + \delta$  increases and  $\delta(A) = 3q^2/4 > a - A$ .

Only safe can be purchased positively. Its value  $y > d$  is the own  $Q$  high coordinate. The reverse  $Q$  menu has scarce-upgrade threshold at least  $k(y) \geq c$ , so its own  $E$  low value  $\rho < c$  cannot obtain the conflicting item as scarce or bundle. This covers the inverse plateau using the weak inequality  $k(y) \geq c$ .

For  $E/E$  with the same physical orientation, an own low value below  $c$  makes bundle and lottery dominated by scarce. Only the larger high value can buy that item; equal high values give zero utility. Under opposite orientations, each scarce value is below  $c$ , the bundle upgrade  $t - q$  and the lottery cell's minimum scarce value  $j > d$ . Only the disjoint own high items can be allocated. These cases exhaust all menu pairs. Closed  $Q$ , open  $E$  and the remaining base rule cover all report boundaries.

#### A.5 Exact integration of the seed

For a fixed opponent report  $(t, \rho)$  on  $\mathcal{T} = \{0 \leq \rho \leq t \leq 1\}$ , each menu option has an affine utility in the own type. Its demand region is the intersection of its utility-comparison half-planes with the unit square. Therefore its payment contribution is its price times that polygon's area; lottery probabilities enter both the half-planes and the item-count integrand. With  $r_c$  as defined in (5),

$$R_* = 4 \int_{\mathcal{T}} r_c(t, \rho) dt d\rho.$$

The factor four is two bidders and two opponent orientations. The integral includes the entire inverse plateau, the strip  $A < t < a$  in  $E$ , and the fee strip  $c \leq \rho \leq b - A$ ,  $A < t < b - \rho$ .

Exact polynomial/simplex integration and a separate decomposition using Green boundary integrals agree on all three algebraic coefficients in (3). An implementation importing neither calculator reconstructs the full integral by vertical slicing and obtains the same coefficients. These are complete evaluations of the stated seed menus; no revenue from a different parameter choice is added. Positive-coordinate tie interfaces have zero own-type area, while the pointwise

argument above defines the mechanism on those interfaces. The package README identifies the structural proof and each independent integral implementation.

## A.6 Certified reserve choice and family boundary

For the fixed seed, the face formula is a polynomial  $R(t)$  of degree at most five. Write it as  $\sum_m mw_m(t)$ , where the twelve fixed moments  $m$  have rational enclosures and  $w_m$  are the explicit polynomial weights in (4). To bound  $R(t_1) - R(t_2)$ , form  $w_m(t_1) - w_m(t_2)$  first and then choose the appropriate endpoint of each moment interval according to that coefficient's sign. This retains the common-moment dependence instead of subtracting two independent revenue intervals. It gives

$$R(83/10000) - R(1/100) > 0.0000021293,$$

while

$$-0.002506136 < R'(1/100) < -0.002504465 < 0.$$

The family optimum can also be localized exactly. If a polynomial on  $[a, b]$  is expanded as  $p(a + (b - a)z) = \sum_{k=0}^n c_k z^k$ , its Bernstein coefficients are

$$b_i = \sum_{k=0}^i c_k \frac{\binom{i}{k}}{\binom{n}{k}}, \quad 0 \leq i \leq n.$$

Each coefficient here is a linear form in the same fixed moments. Exact interval bounds for every coefficient prove

$$\begin{aligned} R'(t) &> 0 & (0 \leq t \leq 0.008297), \\ R'(t) &< 0 & (0.008299 \leq t \leq 1), \\ R''(t) &< 0 & (0.008297 \leq t \leq 0.008299). \end{aligned}$$

The polynomial extends continuously to  $R(1) = 0$ , although the mechanism is defined only for  $t < 1$ . The derivative signs and strict concavity on the middle interval imply a unique global family maximizer

$$0.008297 < \tau_* < 0.008299.$$

Bernstein bounds on that middle interval also give  $\max_{0 \leq t < 1} R(t) < 0.87651436$ . This is a fixed-seed family ceiling, not an unrestricted auction upper. The selected simple rational  $\tau = 83/10000$  is near the family maximizer and is not asserted to be family-optimal or globally optimal.

The complete outward-rounded endpoints used by the release are

$$\begin{aligned} R_\tau &\in [0.876514341027067549900397423113, \\ &\quad 0.876514355424052828335888359874], \\ U &\in [0.882351585488796576205586892263, \\ &\quad 0.882351585488796576205586892264], \\ U - R_\tau &\in [0.005837230064743747869698532390, \\ &\quad 0.005837244461729026305189469151]. \end{aligned}$$

The endpoint strings record rational bounds; their printed length is not the precision of the unknown revenue. The parameter checker uses only standard-library rational arithmetic and imports no face calculator. Its input hashes, all Bernstein coefficient intervals, exact gain and updated ownership-box margins are in the reserve-parameter manifest. Finite implementation checks cover reserve faces, inverse knots and menu boundaries separately from the all-real preservation theorem.



## B The radial identity and rational stream field

### B.1 Analytic weak duality

For own type  $v_i = (x, y)$  and  $r = \max(x, y)$ , define the radial field

$$F^0(x, y) = \frac{3r^2 - 1}{2r^2}(x, y), \quad F^0(0, 0) = 0.$$

Its value at the origin is immaterial to integration. Since  $|F^0| = O(1/r)$ , the field is integrable on the square.

**Theorem 7** (Stream weak duality). *For each bidder  $i$ , let  $G_i$  be a jointly  $C^1$  vector field on  $[0, 1]^4$ . Suppose that, for each opponent report, its own-type divergence is zero and its normal component vanishes on every side of the own-type square. Put  $F_i(v) = F^0(v_i) + G_i(v)$ . Every admissible mechanism  $M$  satisfies*

$$\text{Rev}(M) \leq \mathcal{E}(F) := \int_{[0,1]^4} \sum_{j=1}^2 \max\{0, F_{1j}(v), F_{2j}(v)\} dv. \quad (11)$$

Consequently  $\text{OPT} \leq \mathcal{E}(F)$ .

*Proof.* Fix bidder  $i$  and an opponent report on a full-measure set for which the conditional expected-payment slice is integrable. Write truthful utility as  $u(v)$ , expected allocation as  $X(v) \in [0, 1]^2$ , and expected payment as  $p(v)$ . Pointwise DSIC gives

$$u(v) \geq u(r) + (v - r) \cdot X(r) \quad \text{for every } v, r.$$

Thus  $u$  is convex and  $X(r) \in \partial u(r)$ . These inequalities and  $0 \leq X_j \leq 1$  also make  $u$  Lipschitz. At every differentiability point,  $X = \nabla u$ . At nonsmooth points we retain the mechanism's given jointly measurable subgradient selection; its pointwise feasibility is part of the model, not inferred from an almost-everywhere gradient.

Joint measurability and the assumed integrability of the payment, together with boundedness of  $v \cdot X$ , make  $u = v \cdot X - p$  measurable and integrable. Thus the conditional identities below apply on this full-measure set of opponent reports and may then be integrated by Fubini.

We retain the origin utility explicitly. For jointly completed-Lebesgue measurable input, measurability of its restriction to a null slice does not follow merely by restriction. Instead, continuity on each own-type fiber gives

$$u_i(0; w) = \lim_{n \rightarrow \infty} n^2 \int_{[0, 1/n]^2} u_i(v; w) dv.$$

Fubini supplies measurable versions of these averages outside a common opponent null set. Completeness makes the actual origin trace measurable as well, including its values on that exceptional set. The Lipschitz bound  $|u_i(v; w) - u_i(0; w)| \leq 2$  then implies  $|u_i(0; w)| \leq \int |u_i(v; w)| dv + 2$ , proving its integrability. Ex-post IR gives  $u_i(0; w) \geq 0$ .

Partition the square into the two radial charts

$$v = s(1, t) \quad \text{and} \quad v = s(t, 1), \quad 0 \leq s, t \leq 1,$$

each with Jacobian  $s$ . On a fixed ray set  $q = (1, t)$  or  $(t, 1)$  and  $g(s) = u(sq)$ . Convexity makes  $g$  absolutely continuous and  $g'(s) = q \cdot X(sq)$  almost everywhere and  $g(0) = u(0)$ . Since  $p(sq) = sg'(s) - g(s)$ ,

$$\int_0^1 p(sq) s ds = \int_0^1 \frac{3s^2 - 1}{2} g'(s) ds - \frac{u(0)}{2}.$$

Combining the two charts gives the conditional revenue identity

$$\int_{[0,1]^2} p(v) dv = \int_{[0,1]^2} F^0(v) \cdot X(v) dv - u(0).$$

The correction is bounded, and  $u \in W^{1,\infty}$ . Weak Green integration, zero own-type divergence, and the zero normal trace therefore give

$$\int_{[0,1]^2} G_i \cdot \nabla u = 0.$$

Thus replacing  $F^0$  by  $F_i$  preserves the conditional identity.

Apply this identity to both bidders and then integrate over opponent reports. Fubini is justified by the integrability established above. Dropping the nonpositive origin terms leaves the sum of the field-allocation pairings. For every profile and item  $j$ , feasibility gives  $X_{1j} + X_{2j} \leq 1$ , with both allocations nonnegative. Hence

$$\sum_i F_{ij} X_{ij} \leq \max\{0, F_{1j}, F_{2j}\}.$$

This pointwise relaxation is valid for arbitrary expected allocations and therefore for arbitrary internal randomization. Integrating proves (11); the maximizing allocation in its integrand need not itself be implementable by a DSIC mechanism.  $\square$

**Boundary interpretation.** The radial identity can also be checked by integration by parts. The classical divergence of  $F^0$  is 3 in each open triangle  $x > y > 0$  and  $y > x > 0$ , hence almost everywhere. Continuity across the diagonal away from the origin makes the interface fluxes cancel. The outer boundary flux is 2, while the flux around a puncture at the corner  $(0,0)$  tends to 1, with the normal pointing into the removed square. Applying Green's formula to the two triangles outside  $[0, \varepsilon]^2$  and letting  $\varepsilon \downarrow 0$  gives

$$\int F^0 \cdot \nabla u = \int_{x=1} u + \int_{y=1} u - 3 \int_{[0,1]^2} u + u(0).$$

Lipschitz continuity replaces the trace at the puncture by  $u(0)$  with vanishing error. This accounts for the corner term without treating a boundary corner as an interior distributional source.

## B.2 The rational stream witness

The correction is an opponent-dependent curl. For an exponent  $e = (e_0, e_1, e_2, e_3)$ , put

$$m_e = x^{e_0} y^{e_1} z^{e_2} w^{e_3}, \quad \sigma(e) = (e_1, e_0, e_3, e_2).$$

Let  $E$  be the lexicographically smaller member of every nontrivial  $\sigma$ -pair of total degree at most four. Define

$$h(x, y, z, w) = \sum_{e \in E} \theta_e (m_e - m_{\sigma(e)}), \quad \psi = x(1-x)y(1-y)h, \quad G = (\partial_y \psi, -\partial_x \psi).$$

The thirty-two rational coefficients are listed below in the exact basis order used by the verifier. A convex quasi-Monte Carlo search was used only to discover them; the proof defines the field

by the following rational numbers and does not trust the numerical optimization.

| $e$          | $\theta_e$      | $e$          | $\theta_e$      |
|--------------|-----------------|--------------|-----------------|
| (0, 0, 0, 1) | −2995385/993184 | (0, 0, 0, 2) | 2337131/676164  |
| (0, 0, 0, 3) | 5724621/964727  | (0, 0, 0, 4) | −4591777/714341 |
| (0, 0, 1, 2) | −834567/846478  | (0, 0, 1, 3) | 6291553/970203  |
| (0, 1, 0, 0) | 2762675/446201  | (0, 1, 0, 1) | 206964/961681   |
| (0, 1, 0, 2) | −6090207/979156 | (0, 1, 0, 3) | 6862090/913753  |
| (0, 1, 1, 0) | −5206994/854575 | (0, 1, 1, 1) | 515889/117356   |
| (0, 1, 1, 2) | −3346363/843816 | (0, 1, 2, 0) | −125263/465506  |
| (0, 1, 2, 1) | 3005651/895053  | (0, 1, 3, 0) | −1043238/468269 |
| (0, 2, 0, 0) | 1660919/994155  | (0, 2, 0, 1) | 1276309/943222  |
| (0, 2, 0, 2) | −2075136/255347 | (0, 2, 1, 0) | 404816/59269    |
| (0, 2, 1, 1) | 250988/352269   | (0, 2, 2, 0) | 3825425/994154  |
| (0, 3, 0, 0) | −5383501/354845 | (0, 3, 0, 1) | 3312081/940771  |
| (0, 3, 1, 0) | −4885471/759831 | (0, 4, 0, 0) | 7648954/974695  |
| (1, 1, 0, 1) | −603823/508896  | (1, 1, 0, 2) | 417332/188831   |
| (1, 2, 0, 0) | 17884893/955196 | (1, 2, 0, 1) | 992961/785407   |
| (1, 2, 1, 0) | 2761171/712822  | (1, 3, 0, 0) | −3841999/611370 |

Give bidder 2 the same construction with  $(x, y)$  and  $(z, w)$  swapped, and put  $F_i = F_i^0 + G_i$ .

**Lemma 8** (Rational witness feasibility). *These fields satisfy Theorem 7. They are equivariant under bidder exchange and simultaneous item exchange.*

*Proof.* The corrections are polynomials. Equality of mixed partial derivatives gives  $\text{div}_{x,y} G = 0$ . Moreover  $G \cdot n = 0$  on each side of the own-type square: on a vertical side  $\partial_y \psi = 0$ , and on a horizontal side  $-\partial_x \psi = 0$ , because of the boundary factors in  $\psi$ . Bidder symmetry follows from the definition of  $G_2$ . Finally,  $h(y, x, w, z) = -h(x, y, z, w)$ , hence the same antisymmetry holds for  $\psi$ . Differentiation gives

$$G(y, x, w, z) = (G_y(x, y, z, w), G_x(x, y, z, w)),$$

where  $G_x, G_y$  denote the two components. The radial field has the same item-exchange property, proving the claim.  $\square$

## C Conditional support and certified integral components

**A reusable conditional inequality.** Set  $A = 2/3$ ,  $b_0 = (4 - \sqrt{2})/3$ ,  $k_0 = b_0 - A$ ,  $Z = 3/4$  and

$$\ell(x) = \begin{cases} A, & 0 \leq x \leq k_0, \\ b_0 - x, & k_0 < x < A, \\ k_0, & A \leq x \leq 1, \end{cases} \quad D_0 = \{x < A, y < \ell(x)\}.$$

Let  $f(x) = 3\ell(x) - 2$  for  $0 \leq x < A$  and  $f(x) = 1$  for  $A \leq x \leq 1$ ,  $F(x) = \int_x^1 f(s)ds$  and  $H_Z(y) = 4(y - Z)_+$ . Define

$$\begin{aligned} \Psi_1(x, y) &= 3(x - A)_+ + 4F(x)\mathbf{1}_{y > Z}, \\ \Psi_2(x, y) &= 3(y - \ell(x))_+ \mathbf{1}_{x < A} + f(x)H_Z(y). \end{aligned} \tag{12}$$

The first term of  $\Psi_2$  is zero for  $x \geq A$ . These formulas fix a Borel version on all boundaries. Both densities are bounded and nonnegative. For example, above  $Z$  and below  $A$  the second

density is  $(1 - H_Z)3(y - \ell) + H_Z(3y - 2) \geq 0$ . The edge-transfer envelope gives the universal identity

$$\langle \Psi, a \rangle - R = 3 \int_{D_0} u \geq 0 \quad (13)$$

for every measurable, integrable randomized DSIC/IR conditional mechanism, including unnormalized utility. Applied to a complete jointly measurable mechanism, it holds on almost every opponent fiber and yields the integrated inequality by Fubini. Equivalently,  $\text{div } \Psi = 3$  outside  $D_0$  and zero inside, with the required top/right boundary flux. The menu  $\max(0, x - A, y - A, x + y - b_0)$  attains equality and saturates every positive price, so  $\int (\Psi_1 + \Psi_2) = R_{\text{SB}} = 4/9 + 2\sqrt{2}/27$ . Identity (13) holds without imposing occupation constraints on the residual.

**Compatibility with the opposite bidder.** Write  $W_0 = [0, 43/100]^2$ , for the support switch. The support  $\Psi$  vanishes on  $W_0$ . For the specified stream field  $\phi_i$ , exact rational Bernstein subdivision on both radial charts proves

$$\phi_{ij}(v_i, v_{-i}) \leq 0 \quad \text{if } v_i \in W_0 \quad (j = 1, 2),$$

for every opposing report. The field value at its integrable radial origin can be set to zero; the origin IR trace remains in the envelope. This sign condition permits replacing prices according to the opponent's report while retaining an upper bound on its own complete screening problem.

Add four symmetric, disjoint two-way IC flows to obtain  $\phi'_i$ . One flow acts for a fixed opposing box centered at  $(3/5, 1/5)$ , joining own-report boxes centered at  $(2/5, 39/40)$  and  $(7/20, 23/40)$ . All coordinate halfwidths are  $3/200$  and the translated-pair flow density is  $1/20$ . Forward and reverse flows have equal mass. Bidder exchange and simultaneous item exchange produce four copies. Their eight profile boxes avoid every profile with either bidder in  $W_0$ . Each weighted inequality is an actual DSIC deviation with the opponent held fixed; mass balance cancels utilities and introduces no interior sink. The virtual correction is opposite on the paired boxes.

Define a common nonnegative capacity density, in physical item coordinates,

$$\Pi'_j(w, v) = \begin{cases} 0, & w, v \in W_0, \\ \Psi_j(v), & w \in W_0, v \notin W_0, \\ \Psi_j(w), & v \in W_0, w \notin W_0, \\ \max(0, \phi'_{1j}, \phi'_{2j}), & w, v \notin W_0. \end{cases} \quad (14)$$

**Theorem 9** (A common unrestricted charged support). *For the full class of complete single-bidder DSIC/IR mechanisms with marginal allocations in  $[0, 1]^2$ , define*

$$H_i(\Pi) = \sup_{M_i} \{\text{Rev}_i(M_i) - \langle \Pi, a_i \rangle\}.$$

*Then  $H_1(\Pi') = H_2(\Pi') = 0$  and  $\text{OPT} \leq \int \sum_j \Pi'_j$ .*

*Proof.* When the opponent belongs to  $W_0$ , use (13). Outside that opponent region, the stream envelope and the added nonnegative IC inequalities bound revenue by the  $\phi'_i$  allocation pairing minus nonnegative IC/IR slack. If the own report is in  $W_0$ , its virtual coefficients are nonpositive and the cycle correction is zero; otherwise (14) dominates them. Thus  $\text{Rev}_i(M_i) - \langle \Pi', a_i \rangle \leq 0$  for every complete  $M_i$ . The empty mechanism attains zero. Integrating joint capacity gives the auction bound. This argument applies directly to jointly measurable mechanisms with integrable payments and bounded allocations; it never interchanges an uncountable supremum with an integral. Positive field parts are integrable, and the absolutely continuous pairings cover the Lebesgue-measurable class stated in Section 2.  $\square$

**Common price and its mass.** The field  $\phi_i = F_i$  of Lemma 8 already defines a common density  $\Pi_j^0 = \max(0, \phi_{1j}, \phi_{2j})$ . The stream identity gives

$$\text{Rev}_i(M_i) = \int \sum_j \phi_{ij} a_{ij} - \int u_i(0; v_{-i}) \leq \langle \Pi^0, a_i \rangle,$$

and the empty mechanism attains equality. Hence  $H_1(\Pi^0) = H_2(\Pi^0) = 0$  as well. Existence of a common measure and zero charged values are already consequences of the specified stream argument. The conditional replacements and compatible IC cycles below instead prove a strict decrease in its total mass,  $\int \Pi' < \int \Pi^0$ , while preserving validity for both unrestricted conditional problems. This integrated comparison does not assert a pointwise price decrease.

Solving both charged problems does not produce an equality auction. For example, empty leaves all positively priced capacity unused. A matching mechanism must satisfy the capacity and incentive equalities at the same time.

**A rational endpoint with componentwise replay.** Let  $B$  be the exact integral of the original stream maximum. The new continuous majorant uses

$$\max \left( 0, \sum_k B_k a_k, \sum_k B_k b_k \right) \leq \sum_k B_k \max(0, a_k, b_k)$$

for nonnegative tensor Bernstein basis functions  $B_k$  summing to one. Integrating the right-hand side retains within-box variation. Exact rational construction, directed fixed-point error radii and complete dyadic coverage give

$$B \leq B_{20} = \frac{231831916327659047}{262144000000000000}.$$

The primary and independently reconstructed traversals each visit 1,351,220 tree nodes and agree on the five recorded depth bounds and all chart coverages.

Let  $D$  be the reduction obtained by the two conditional replacements before the cycles. The overlap  $W_0 \times W_0$  has zero old and new price, so the two reductions add. At the single-buyer menu  $a^*$ ,

$$D = 2 \int_{w \in W_0} \int_v \sum_j [(\phi_{2j}(v, w))_+ - a_j^*(v) \phi_{2j}(v, w)].$$

The integrand is nonnegative. Integrate opponent variables exactly and partition each own radial chart into rational rectangles wholly inside a single menu region. On each such rectangle  $K$  use  $\int_K (f)_+ \geq (\int_K f)_+$ . Boundary-crossing rectangles are omitted. In particular, a bundle rectangle must meet both the total-price and lower-coordinate upgrade thresholds. Requiring total value alone would incorrectly include some singleton buyers. Seven nested partitions give increasing rational lower bounds  $\Delta_{16}, \dots, \Delta_{1024} \leq D$ . These partitions certify integrals over continuous types; they do not replace the auction domain by a grid. The complete 1024-partition accumulation is performed by the primary verifier. The independent check reconstructs all averaged coefficients, cross-checks classifications at sizes up to 64 and selected cell integrals, and reuses the primary classifier and integral tables. It does not perform a second complete 1024-partition accumulation.

On every sparse-cycle box, exact polynomial bounds keep the old and new virtual winners strict. Integration then gives the exact combined reduction  $81/10^{10}$ . Disjointness from the conditional replacement regions makes the two improvements compatible. Define the reference accounting constant

$$U_{\text{ref}} = B_{20} - \Delta_{1024} - \frac{81}{10^{10}}, \quad \text{OPT} \leq U_{\text{ref}}. \quad (15)$$

Here  $\Delta_{1024}$  is the final exact rational `delta` in `conditional_global_splice.json`; its full numerator and denominator and those of  $U_{\text{ref}}$  are exported in the coordinated component manifest.

**Which equality conditions remain unsatisfied.** Put  $O_i = \{v_{-i} \in W_0\}$ . The exact gap for the measure is the sum of four nonnegative mechanism slacks,

$$\begin{aligned} S_{\text{screen}} &= \sum_i 3 \int_{O_i} \int_{D_0} u_i, \\ S_{\text{capacity}} &= \sum_j \int \Pi'_j (1 - a_{1j} - a_{2j}), \\ S_{\text{virtual}} &= \sum_{i,j} \int_{O_i^c} (\Pi'_j - \phi'_{ij}) a_{ij}, \\ S_{\text{IC}} &= \sum_i \int_{O_i^c} u_i(0; v_{-i}) + \sum_{\text{cycle edges}} \int \text{weighted truthful IC residual}. \end{aligned}$$

They satisfy  $\int \Pi' - R = \sum S$ . The origin IR trace is explicit; there is no positive interior sink at every type. Screening's internal utility terms are not counted again in  $S_{\text{IC}}$ . The reported rational endpoint additionally includes

$$E_{\text{cert}} = (B_{20} - B) + (D - \Delta_{1024}) \geq 0, \quad U_{\text{ref}} - R = \sum S + E_{\text{cert}}.$$

Numerical majorization slack is thus separated from mechanism equality. The final certificate adds the terms in Section 4.3. The reference decomposition above identifies the numerical remainder used there, while the final measure has its own source, incentive and capacity slacks. Neither decomposition alone supplies a matching allocation.

### C.1 The overlap-safe support splice

Let  $K = J \cup J^{\text{swap}}$ ,  $D^\sharp = W_0 \cup K$ , and let  $\Pi^W$  be the common price with only the  $W_0$  switch. The added rectangles satisfy  $\max(w) \leq 1/2$  and  $\sum_j w_j \leq 17/20 < b_0$ ; also  $2(43/100) < b_0$ . Thus both orientations of  $\Psi$  vanish on  $D^\sharp \subset D_0$ . All IC corrections used in the final certificate avoid profiles where either bidder belongs to  $D^\sharp$ .

**Lemma 10** (Complete splice decrement). *For  $L, E$  and  $J$  as in (9),*

$$\int \sum_j (\Pi_j^W - \Pi_j) \geq 4[L - |J|R_{\text{SB}} - E], \quad |J| = 49/2000,$$

*with the same inequality after adding the disjoint IC corrections.*

*Proof.* For one oriented added opponent  $w \in J$ , set  $a = \phi_j(v; w)$ ,  $b = \phi_j(w; v)$  and  $p = \Psi_j(v) \geq 0$ . If  $v \notin D^\sharp$ , the prices are  $\max(0, a, b)$  and  $\max(p, b)$ , so their difference is at least  $a_+ - p - (b - p)_+$ .

If  $v \in W_0$ , both actual prices are zero. The low-square sign certificate gives  $a \leq 0$ , and both supports vanish. The directed expression equals  $-b_+ \leq 0$ , so it is still a valid lower bound.

If both reports belong to  $K$ , both replacement supports vanish. The two directed expressions, using the respective support orientations, are  $a_+ - b_+$  and  $b_+ - a_+$ . Their sum is zero, while the actual old price is nonnegative and the new price is zero. The two replacements are therefore charged together, without assuming disjoint bidder events.

If neither report belongs to  $K$ , neither the price nor the directed sum changes. These cases exhaust the report space, including cross-orientation overlaps. Integration of each directed expression gives  $L - |J|R_{\text{SB}} - E$ . Bidder exchange and the two oriented rectangles give factor four; the item sum is already inside  $L$  and  $E$ . Disjoint IC supports are unaffected by every switch, so their certified decrements add separately. This proves the claim.  $\square$

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